

struction set architecture (ISA) based on established reduced instruction selected computing (RISC) principles. Unlike proprietary ISAs, RISC-V is free to implement and is not subject to licensing fees, making it an attractive option for hardware developers looking to customise their processors for specific applications. It allows for a wide range of implementations, from tiny microcontrollers to high-performance servers, and supports a growing ecosystem of software and tools.

Fabricate for chiplet flow.

Chiplets are a novel approach to semiconductor design and manufacturing that involves creating minor, modular blocks of integrated circuits, each with a specific function, which get combined on a single package to form a complete system. It contrasts with the traditional monolithic approach, where all components are fabricated on a single silicon die. Chiplets enable greater flexibility and scalability, allowing chip architectures to meet the requirements of a specific application. Chiplets also allow for more efficient use of silicon area while enabling the use of different process nodes, thus lowering the cost of production.

Packaging/testing with heterogeneous approach. Heterogeneous integration refers to the advanced manufacturing technique in the semiconductor industry where diverse and separately manufactured components such as chiplets, memory, analog, digital, and photonic devices are integrated into a single system or package. This approach enables combining components

that may be built using different materials and process technologies, optimising performance and functionality that cannot be achieved with homogeneous (monolithic) integration. Heterogeneous integration allows for greater design flexibility, improved performance through specialisation, and potential cost savings from yield enhancements and mature fabrication processes.

SoC development must always focus on utilising novel techniques that can defy the traditional way of SoC development to provide chip architects with more room to pack new features (like NPUs, different ASICs, eFPGAs, and so on).

Thus, MtM with RISC-V, chiplet, and heterogeneous integration can potentially revolutionise the SoC industry (mainly different types of XPU), which is also at the core of

the AI revolution. The more advanced the AI implementation is, the more influential the underlying SoC architecture should be. MtM's way of designing and developing SoC is the way to enable such high-performance architecture.

Future of More-than-Moore SoCs

The semiconductor industry (mainly the companies working on advanced SoCs) knows the benefits of the MtM solutions available today, but the rate of adoption could be much higher. Essentially, this is due to the ecosystem around RISC-V, chiplet, and heterogeneous integration, which is still under develop-

ment. It will take about half a decade before the software-to-silicon ecosystem can enable SoC vendors to utilise these three breakthroughs efficiently.

Currently, the use of chiplet and heterogeneous integration is growing, with established CPU and GPU vendors leading the race. As more global emerging startups utilise the openness of RISC-V with the flexibility of chiplet and heterogeneous integration, the avenues to develop better AI-friendly SoCs will open, unlocking the market to speed up the adoption of such flow of SoC development.

Until then, the semiconductor industry is expected to follow Moore's way of designing and manufacturing SoCs with the More-than-Moore approach right behind it to enable more cost, node, and feature-friendly solutions for 1nm and beyond technology nodes. **EFY**

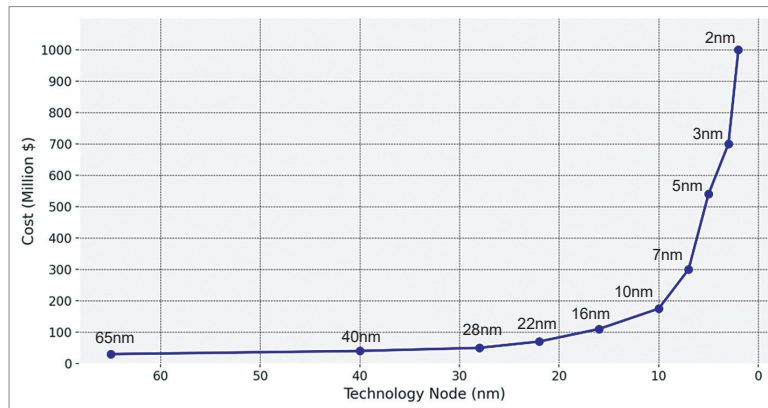


Fig. 2: SoC production cost based on technology node

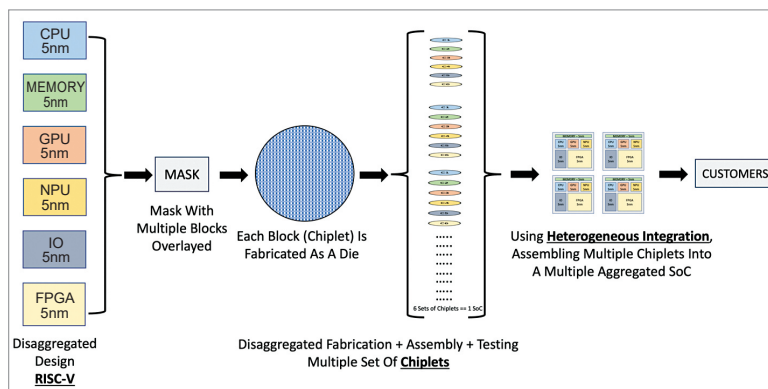


Fig. 3: SoC development—More than Moore way